EE656: Robust Control
Model answer Quiz 1

`Set - A`

Q1: Prove or disprove the following:

(a) If an observer exists, the system is observable.
Solution: ***False*** *(Detectability is a necessary and sufficient condition)*
Counter example to the statement:

$$\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

This is detectable. Consider $L = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$\sigma(A + LC) = \sigma\left(\begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right) = \sigma\left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \right) \subseteq \mathbb{C}^-$$

Thus, the observer -

$$\dot{\hat{x}} = A\hat{x} + Bu(t) + L(\hat{y} - y) \quad \text{and} \quad \hat{y} = C\hat{x} + Du(t)$$

will result in

$$\lim_{t \to \infty} [x(t) - \hat{x}(t)] = 0$$

(b) Every invariant zero is a transmission zero of the system.
Solution: ***False*** (This is true when (A,B,C,D) is a minimal realization of G(s))
e.g. $G(s) = \dfrac{s+2}{s+1}$ and consider $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = 1.$
You can verify that $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$
From $G(s) = \dfrac{s+2}{s+1}$, it is clear that $\{-2\}$ is a transmission zero.
Now, let $\lambda = -1$.

$$rank \begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} = \texttt{rank} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = 2 < \texttt{nrank} \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = 3$$

$\therefore -1$ is an invariant zero but not a transmission zero.

(c) Given an uncontrollable system, there exists an initial condition such that the system can be brought to rest in finite time.
Solution: ***False*** (Input needs to present for atleast some control)
e.g. $\dot{x} = x$: This is an uncontrollable system (more precisely autonomous). The states are $x(t) = e^t x(0)$. Clearly, $\nexists$ initial condition which can be brought to rest.

(d) The system (A, B) is controllable if and only if $(A + 2I, B)$ is controllable.
Solution: ***True***
If: $(A + 2I, B)$ controllable $\Rightarrow$ (A, B) controllable.
We prove the contrapositive statement, i.e., (A, B) uncontrollable $\Rightarrow$ $(A + 2I, B)$ uncontrollable.

$\because$ (A, B) uncontrollable, $\exists \mu \in \mathbb{C}$ and $\mu \in \sigma(A)$ such that

$$Av = \mu v \iff (A + 2I)v = (\mu + 2)v \tag{1}$$

Now conducting PBH test at $(\mu + 2)$ eigenvalue of $(A + 2I)$.

$$v^* \begin{bmatrix} (A + 2I) - (\mu + 2)I & B \end{bmatrix} = 0 \qquad \text{(using eq 1)}$$

$\therefore$ $(\mu + 2)$ is an uncontrollable eigenvalue of $(A + 2I)$ and thus, $(A + 2I, B)$ is uncontrollable.
Only if: (A,B) controllable $\Rightarrow$ (A+2I, B) controllable.
We prove: (A+2I, B) uncontrollable $\Rightarrow$ (A,B) uncontrollable.

Let μ be an uncontrollable eigenvalue of $(A + 2I, B)$ and $(A + 2I)v = \mu v \Rightarrow Av = (\mu - 2)v$. Hence, PBH test on (A,B) at $\lambda = \mu - 2$ gives:

$$v^* \begin{bmatrix} A - (\mu - 2)I & B \end{bmatrix} = v^* \begin{bmatrix} (A + 2I) - \mu I & B \end{bmatrix} = 0$$

$\therefore$ (A,B) is uncontrollable.

(e) $z_0 \in \mathbb{C}$ is an invariant zero of the system (A, B, C, D) if and only if there exists a non-zero $x \in \mathbb{C}^{\mathtt{n}}$ and $u \in \mathbb{C}^{\mathtt{m}}$ such that

$$\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = 0.$$

Solution: ***False*** $\left(\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \text{have to be full column normal rank.} \right)$
Let, $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \end{bmatrix}, D = 0$
Then,

$$\begin{bmatrix} 1 - z_0 & 0 & 1 & -1 \\ 0 & 1 - z_0 & -1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = 0 \qquad \text{for any } z_0 \in \mathbb{C}$$

Thus, according to the statement any $z_0 \in \mathbb{C}$ is an invariant zero. (Not possible).

Q2: Given the system

$$\dot{x} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -2 & -1 & 7 & -7 \\ -1 & 0 & 12 & 1 & 9 & 1 \\ -2 & -3 & -7 & -3 & -9 & 2 \\ 4 & 5 & 15 & 1 & 21 & -4 \\ 1 & 2 & 0 & 3 & 1 & -1 \\ 2 & 3 & 1 & 3 & 9 & -8. \end{bmatrix} x \qquad y = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{bmatrix} x.$$

Design an observer with poles at $\{-1, -1, -1, -2, -2, -2\}$. In case you are unable to place all the poles to the desired location, design the observer such that you can place as many poles as possible at the desired location.

Solution: Let $A_1 = A^T$ and $B_1 = C^T$. Note that $\texttt{rank}(\mathcal{C}_{A_1B_1}) = 3 < 6$. Therefore, the system is uncontrollable and dimension of controllable space is 3. We take a basis of the controllability space and extend it to $\mathbb{R}^6$.

On computation, we have $\mathcal{C}_{A_1B_1} = \langle \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 4 \\ -1 \end{bmatrix} \rangle$.

We define $T := \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 4 & 0 & 0 & 0 \\ -1 & 1 & -1 & 0 & 0 & 0 \end{bmatrix}$. Using T as the similarity transformation matrix, we get

$$\hat{A} = T^{-1}AT = \left[\begin{array}{ccc|ccc} 0 & 0 & -1 & 11/2 & 5/2 & 3/2 \\ 0 & -3 & 0 & 3/2 & 7/2 & 1 \\ 1 & 0 & 2 & -1/2 & 1/2 & -3/2 \\ \hline 0 & 0 & 0 & -9/2 & -7/2 & -1 \\ 0 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 0 & 0 & 4 & 4 & 0 \end{array}\right] =: \left[\begin{array}{c|c} A_c & \hat{A}_{12} \\ \hline 0 & A_{uc} \end{array}\right],$$

$$\hat{B} = T^{-1}B = \left[\begin{array}{cc} 1/2 & 0 \\ 0 & 1 \\ 0 & 0 \\ \hline 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] =: \left[\begin{array}{c} B_c \\ \hline 0 \end{array}\right].$$

Note that $\sigma(A_c) = \{1, 1, -3\}$ is the set of controllable eigenvalues and $\sigma(A_{uc}) = \{-2, -2, -1\}$ is the set of uncontrollable eigenvalues.

Desired eigenvalues: $\{-1, -1, -1, -2, -2, -2\}$

We just have to place the controllable modes at $\{-1, -1, -2\}$. Hence, we work with (A_c, B_c) next.

$$A_c = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -3 & 0 \\ 1 & 0 & 2 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} =: \begin{bmatrix} b_1 & b_2 \end{bmatrix}$$

Step 1: $L = \begin{bmatrix} b_1 & Ab_1 & b_2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix}$. Here $d_1 = 2, \quad d_2 = 1, \quad \sigma_1 = 2, \quad \sigma_2 = 3.$

step 2: $L^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}$. Therefore, $q_1 = \begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$, $q_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$.

step 3: $Q = \begin{bmatrix} 0 & 0 & 2 \\ 2 & 0 & 4 \\ 0 & 1 & 0 \end{bmatrix}$. Using Q for another similarity transform.

$$\therefore \hat{A} = QA_cQ^{-1} = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right], \qquad \hat{B} = QB = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right]$$

Choose $\hat{F} = \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \in \mathbb{R}^{2\times 3}$.

$$\begin{aligned} \therefore \hat{A} + \hat{B}\hat{F} &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right] + \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right] \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \\ &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1+f_1 & 2+f_2 & 0 \\ \hline 0 & 0 & -3+f_3 \end{array}\right] \end{aligned}$$

If we choose $f_3 = 1$ then one of the eigenvalues of $\hat{A} + \hat{B}\hat{F}$ becomes -2. Next, we choose f_1 and f_2 such that the other two eigenvalues become -1,-1.

$\because (s+1)^2 = s^2 + 2s + 1, \quad 2 + f_2 = -2 \Rightarrow f_2 = -4$ and $-1 + f_1 = -1 \Rightarrow f_1 = 0.$

$$\begin{aligned} \therefore \hat{F} &= \begin{bmatrix} 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \therefore F_c &= \hat{F}Q = \begin{bmatrix} -8 & 0 & -16 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

We can take $\hat{F} = \begin{bmatrix} F_c & M \end{bmatrix}$, where $M \in \mathbb{R}^{2\times 3}$ and can be choosen arbitrarily. We chose $M = 0$. The feedback matrix is, therefore,

$$F_1 = \hat{F}T^{-1} = \begin{bmatrix} 0 & 0 & 0 & 16 & -8 & 0 \\ 0 & 0 & 0 & -3 & 1 & 1 \end{bmatrix}$$

(one can check that $\sigma(A_1 + B_1F_1) = \{-1, -1, -1, -2, -2, -2\}$)
Using duality of controllability and observability, we have

$$L = F_1^T = \begin{bmatrix} 0 & 0 & 0 & 16 & -8 & 0 \\ 0 & 0 & 0 & -3 & 1 & 1 \end{bmatrix}^T \quad \text{and} \quad \sigma(A+LC) = \{-1, -1, -1, -2, -2, -2\}.$$

Q3: Consider a system with the following transfer function:

$$G(s) = \begin{bmatrix} \frac{s-1}{s+1} & \frac{s-1}{(s+1)^2} \\ 0 & \frac{s+2}{s^2-1} \end{bmatrix}$$

(a) Find the transmission zeros of the system.

(b) Find a minimal realization of the system and ascertain the invariant zeros of the system

(c) Find the poles of the system.

Solution: We find the McMillan form first.

$$G(s) = \begin{bmatrix} \dfrac{s-1}{s+1} & \dfrac{s-1}{(s+1)^2} \\ 0 & \dfrac{s+2}{s^2-1} \end{bmatrix} = \frac{1}{(s+1)^2(s-1)} \underbrace{\begin{bmatrix} (s-1)^2(s+1) & (s-1)^2 \\ 0 & (s+2)(s+1) \end{bmatrix}}_{P(s)}.$$

$U(s)$	$P(s)$	$V(s)$	
	$\begin{bmatrix} s^3-s^2-s+1 & s^2-2s+1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ -s & 1 \end{bmatrix}$	$\begin{bmatrix} s^2-2s+1 & s^2-2s+1 \\ 0 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} s^2-2s+1 & s^2-2s+1 \\ 0 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} s^2-2s+1 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$
	$\begin{bmatrix} s^2-2s+1 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 5 & 0 \\ s & 1 \end{bmatrix}$	$\begin{bmatrix} -11s+5 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$
	$\begin{bmatrix} -11s+5 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 5 & 0 \\ -11 & 1 \end{bmatrix}$	$\begin{bmatrix} 36 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$
	$\begin{bmatrix} 36 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 1/36 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$
	$\begin{bmatrix} 1 & -5s-1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 1 & 5s \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & -1 \\ 0 & s^2+3s+2 \end{bmatrix}$
	$\begin{bmatrix} 1 & -1 \\ 0 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & s^2+3s+2 \end{bmatrix}$

$$G_{\mathtt{Mc}}(s) = \frac{1}{(s+1)^2(s-1)} \begin{bmatrix} 1 & 0 \\ 0 & s^2+3s+2 \end{bmatrix} = \begin{bmatrix} \dfrac{1}{(s+1)^2(s-1)} & 0 \\ 0 & \dfrac{s+2}{(s+1)(s-1)} \end{bmatrix}$$

a) The set of transmission zeroes of $G(s)$: $\{-2\}$.

b) We find the minimal representation next.

$$G(s) = \begin{bmatrix} \frac{s-1}{s+1} & \frac{s-1}{(s+1)^2} \\ 0 & \frac{s+2}{s^2-1} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{s-1}{s+1} & \frac{s-1}{s+1} \\ 0 & \frac{s+2}{s-1} \end{bmatrix}}_{G_1(s)} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{s+1} \end{bmatrix}}_{G_2(s)}$$

We find a minimal realization for $G_1(s)$ first.

$$G_1(s) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} \frac{-2}{s+1} & \frac{-2}{s+1} \\ 0 & \frac{3}{s-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \frac{1}{s+1} \begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix} + \frac{1}{s-1} \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$$

$$A_1 = \left[\begin{array}{c|c} -1 & 0 \\ \hline 0 & 1 \end{array}\right], \qquad B_1 = \left[\begin{array}{cc} -2 & -2 \\ \hline 0 & 3 \end{array}\right], \qquad C_1 = \left[\begin{array}{c|c} 1 & 0 \\ 0 & 1 \end{array}\right], \qquad D_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Next, we find a minimal realization for $G_2(s)$.

$$G_2(s) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{s+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{s+1}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_2 = -1, \qquad B_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad C_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad D_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\therefore G_1(s)G_2(s)$ has the following realization:

$$\begin{aligned} A &= \begin{bmatrix} A_1 & B_1C_2 \\ 0 & A_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & -1 \end{bmatrix}, B = \begin{bmatrix} B_1D_2 \\ B_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \\ C &= \begin{bmatrix} C_1 & D_2C_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, D = D_1D_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$\because$ $\texttt{rank}(\mathcal{C}_{AB}) = 3$ and $\texttt{rank}(\mathcal{O}_{CA}) = 3$, we have a minimal realization of $G(s)$: (A, B, C, D).
Invariant zeros: $\because$ (A, B, C, D) is a minimal realization, the transmission zeros and invariant zeros are same.
$\therefore$ -2 is the invariant zero.

c) $\because$ Eigenvalues of A are $-1, -1, 1$ and (A, B, C, D) is a minimal realization, poles are $\{-1, -1, 1\}$.

Q4: Consider the system given by

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \; y = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

(a) Determine $x(0)$ so that for $u(t) = e^{-4t}, y(t) = ke^{-4t}$, where k is a real constant. Determine k for the present case.
Solution: Using the fact that $\dot{x_1} = x_1 + x_2$, $y = x_1 = ke^{-4t}$, and $u = e^{-4t}$, we find x_2 first.

$$\dot{x_2} = -x_1 + u \Rightarrow \dot{x_2} = -ke^{-4t} + e^{-4t} \Rightarrow x_2(t) = \frac{(k-1)}{4}e^{-4t} + c_1 \qquad (c_1 \text{ is a contant}).$$

Therefore, $\dot{x_1} = -ke^{-4t} + \dfrac{(k-1)}{4}e^{-4t} + c_1 = \dfrac{(-3k-1)}{4}e^{-4t} + c_1$

$$\therefore x_1(t) = \frac{(-3k-1)}{4}\frac{e^{-4t}}{-4} + c_1t + c_2 = \frac{(3k+1)}{16}e^{-4t} + c_1t + c_2 \qquad (c_2 \text{ is a contant}) \tag{2}$$

$\because y(t) = x_1(t)$, we have

$$ke^{-4t} = \frac{(3k+1)}{16}e^{-4t} + c_1t + c_2 \Rightarrow c_1 = 0, c_2 = 0, \text{ and } \frac{(3k+1)}{16} = k.$$

Thus, $k = \dfrac{1}{13}$. Now,

$$x_1(0) = k = \frac{1}{13}, \qquad x_2(0) = \frac{(\frac{1}{13} - 1)}{4} = \frac{-3}{13} \qquad \text{(from (2))}$$

$\therefore$ Initial condition is $\begin{pmatrix} \frac{1}{13} \\ -\frac{3}{13} \end{pmatrix}$ and $k = \frac{1}{13}$.

(b) Let $u(t) = e^{\alpha t}$, Determine $x(0)$ that will result in $y(t) = ke^{\alpha t}$, Determine the conditions on α for this to be true. What is k in this case?

Solution: For $u(t) = e^{\alpha t}$ and $y(t) = ke^{\alpha t}$

$$x_2(t) = \frac{(1-k)}{\alpha} e^{\alpha t} + c_1$$

$$\dot{x_1} = -ke^{\alpha t} + \frac{(1-k)}{\alpha} e^{\alpha t} + c_1 = \left(\frac{-\alpha k + 1 - k}{\alpha} \right) e^{\alpha t} + c_1$$

$$\therefore x_1(t) = \left(\frac{1 - \alpha k - k}{\alpha} \right) \frac{e^{\alpha t}}{\alpha} + c_1 t + c_2 = ke^{\alpha t}$$

$$\therefore c_1 = 0, \qquad c_2 = 0, \qquad 1 - \alpha k - k = \alpha^2 k \Rightarrow k = \frac{1}{\alpha^2 + \alpha + 1}$$

$$\therefore x_1(0) = \frac{1}{\alpha^2 + \alpha + 1} \qquad \& \qquad x_2(0) = \frac{1}{\alpha} \left(1 - \frac{1}{\alpha^2 + \alpha + 1} \right) = \frac{\alpha + 1}{\alpha^2 + \alpha + 1}.$$

Q5: Form a matrix A with your Roll number in the following way:
For example: if your Roll number is 123456789, then the matrix will be of the form:

$$A := \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) Find the diagonal or the Jordan form (whichever is applicable) of A. Name this matrix J.

(b) Construct a matrix B with the minimum number of columns possible such that (J, B) is controllable.

(c) Construct a matrix C with the minimum number of rows possible such that (J, C) is observable.

Solution: This is unique for everyone. No particular answer.

EE656: Robust Control
Model answer Quiz 1

`Set - B`

Q1: Prove or disprove the following:

(a) Every transmission zero is an invariant zero of the system.
Solution: ***True*** Let z_0 be a transmission zero. Hence,

$$\texttt{rank}(G(z_0)) < \texttt{nrank}(G(s)) \tag{1}$$

Note that $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ C(sI - A)^{-1} & I \end{bmatrix} \begin{bmatrix} A - sI & 0 \\ 0 & G(s) \end{bmatrix}$.

$$\texttt{nrank}\left(\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}\right) = \texttt{nrank}(A - sI) + \texttt{nrank}(G(s)) = \texttt{n} + \texttt{nrank}(G(s))$$

Further, at $s = z_0$, $\texttt{rank}\left(\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix}\right) = \texttt{n} + \texttt{rank}(G(z_0))$.

$$\begin{aligned} \because \texttt{n} + \texttt{rank}(G(z_0)) &< \texttt{n} + \texttt{rank}(G(s)) \quad \text{(by eq (1))} \\ \Rightarrow \texttt{rank}\left(\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix}\right) &< \texttt{nrank}\left(\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}\right) \end{aligned}$$

$\therefore z_0$ is an invariant zero.

(b) An observer exists only if the system is observable.
Solution: The statement is: An observer exists *only if* the system is observable. This is the same as if an observer exists, the system is observable. Hence,the statement is ***False*** *(Detectability is a necessary and sufficient condition)*
Counter example to the statement:

$$\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

This is detectable. Consider $L = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$\sigma(A + LC) = \sigma\left(\begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix}\right) = \sigma\left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\right) \subseteq \mathbb{C}^-$$

Thus, the observer -

$$\dot{\hat{x}} = A\hat{x} + Bu(t) + L(\hat{y} - y) \quad \text{and} \quad \hat{y} = C\hat{x} + Du(t)$$

will result in

$$\lim_{t \to \infty} [x(t) - \hat{x}(t)] = 0$$

(c) Given an uncontrollable system there exists an initial condition such that the system can be brought to rest in finite time.
Solution: ***False*** (Input needs to present for atleast some control)
e.g. $\dot{x} = x$: This is an uncontrollable system (more precisely autonomous). The states are $x(t) = e^t x(0)$. Clearly, $\nexists$ initial condition which can be brought to rest.

(d) $z_0 \in \mathbb{C}$ is an invariant zero of the system (A, B, C, D) if and only if there exists a non-zero $x \in \mathbb{C}^{\mathtt{n}}$ and $u \in \mathbb{C}^{\mathtt{m}}$ such that

$$\begin{bmatrix} x^* & u^* \end{bmatrix} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} = 0.$$

Solution: ***False*** $\left(\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \text{ have to be full row normal rank.} \right)$
Let, $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, D = 0$
Then,

$$\begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 - z_0 & 0 & 1 \\ 0 & 1 - z_0 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix} = 0 \qquad \text{for any } z_0 \in \mathbb{C}$$

Thus, according to the statement any $z_0 \in \mathbb{C}$ is an invariant zero. (Not possible).

(e) The system (A, B) is controllable if and only if $(A + \alpha I, B)$ is controllable.
Solution: ***True***

If: (A+αI, B) controllable $\Rightarrow$ (A,B) controllable
We prove the contrapositive statement: If (A,B) uncontrollable $\Rightarrow$ (A+αI, B) uncontrollable.
$\because$ (A,B) uncontrollable. $\exists \mu \in \mathbb{C}$ and $\mu \in \sigma(A)$
such that

$$Av = \mu v \iff (A + \alpha I)v = (\mu + \alpha)v \tag{2}$$

Now conducting PBH test at $(\mu + \alpha)$ eigenvalue of (A+α I).

$$v^* \begin{bmatrix} (A + \alpha I) - (\mu + \alpha)I & B \end{bmatrix} = 0 \qquad \text{(using eq (1))}$$

$\therefore$ $(\mu + \alpha)$ is an uncontrollable eigenvalue of (A+α I).
$\therefore$ (A+α I,B) is uncontrollable.

only if: (A,B) controllable $\Rightarrow$ (A+αI, B) controllable.
We prove: (A+αI, B) uncontrollable $\Rightarrow$ (A,B) uncontrollable.
Let μ be an uncontrollable eigenvalue of $(A + \alpha I, B)$
and $(A + \alpha I)v = \mu v \Rightarrow Av = (\mu - \alpha)v$.
Hence PBH test on (A,B) at $\lambda = \mu - \alpha$ gives-

$$v^* \begin{bmatrix} A - (\mu - \alpha)I & B \end{bmatrix} = v^* \begin{bmatrix} (A + \alpha I) - \mu I & B \end{bmatrix} = 0$$

$\therefore$ (A,B) is uncontrollable.

$$\dot{x} = \frac{1}{9}\begin{bmatrix} 18 & 0 & -90 & -27 & 0 & -45 \\ -11 & 3 & 86 & 13 & 54 & 11 \\ -6 & -9 & -12 & -6 & -18 & 6 \\ 30 & 18 & -3 & -15 & 63 & -30 \\ -1 & 6 & -2 & 11 & -9 & 1 \\ 12 & 9 & -57 & -6 & 0 & -39 \end{bmatrix} x \qquad y = \frac{1}{3}\begin{bmatrix} 2 & 0 & -5 & -1 & 0 & -2 \\ -1 & 0 & 10 & 2 & 3 & 4 \end{bmatrix} x.$$

Design an observer with poles at $\{-1, -1, -1, -2, -2, -2\}$. In case you are unable to place all the poles to the desired location, design the observer such that you can place as many poles as possible at the desired location.

Solution: Let $A_1 = A^T$ and $B_1 = C^T$. Note that $\texttt{rank}(\mathcal{C}_{A_1B_1}) = 3 < 6$. Therefore, the system is uncontrollable and dimension of controllable space is 3. We take a basis of the controllability space and extend it to $\mathbb{R}^6$.

On computation, we have $\mathcal{C}_{A_1B_1} = \langle \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 10 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \\ -1 \\ 1 \\ 9 \\ -4 \end{bmatrix} \rangle$.

We define $T := \begin{bmatrix} 0 & -1 & 4 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 3 & 10 & -1 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 3 & 3 & 9 & 0 & 0 & 0 \\ 0 & 4 & -4 & 0 & 0 & 0 \end{bmatrix}$. Using T as the similarity transformation matrix, we get

$$\hat{A} = T^{-1}AT = \left[\begin{array}{ccc|ccc} 3/2 & 0 & 9/2 & -23/4 & -7/4 & -3/2 \\ 0 & -3 & 0 & 1/2 & 7/6 & 1/3 \\ -1/18 & 0 & 1/2 & 7/4 & 31/36 & 1/6 \\ \hline 0 & 0 & 0 & -9/2 & -7/2 & -1 \\ 0 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 0 & 0 & 4 & 4 & 0 \end{array}\right] =: \left[\begin{array}{c|c} A_c & \hat{A}_{12} \\ \hline 0 & A_{uc} \end{array}\right],$$

$$\hat{B} = T^{-1}B = \left[\begin{array}{cc} -1/2 & 0 \\ 0 & 1/3 \\ 1/6 & 0 \\ \hline 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] =: \left[\begin{array}{c} B_c \\ \hline 0 \end{array}\right].$$

Note that $\sigma(A_c) = \{1, 1, -3\}$ is the set of controllable eigenvalues and $\sigma(A_{uc}) = \{-2, -2, -1\}$ is the set of uncontrollable eigenvalues.

Desired eigenvalues: $\{-1, -1, -1, -2, -2, -2\}$

We just have to place the controllable modes at $\{-1, -1, -2\}$. Hence, we work with (A_c, B_c) next.

$$A_c = \begin{bmatrix} 3/2 & 0 & 9/2 \\ 0 & -3 & 0 \\ -1/18 & 0 & 1/2 \end{bmatrix}, \qquad B_c = \begin{bmatrix} -1/2 & 0 \\ 0 & 1/3 \\ 1/6 & 0 \end{bmatrix} =: \begin{bmatrix} b_1 & b_2 \end{bmatrix}$$

Step 1: $L = \begin{bmatrix} b_1 & Ab_1 & b_2 \end{bmatrix} = \begin{bmatrix} -1/2 & 0 & 0 \\ 0 & 0 & 1/3 \\ 1/6 & 1/9 & 0 \end{bmatrix}$. Here $d_1 = 2, \quad d_2 = 1, \quad \sigma_1 = 2, \quad \sigma_2 = 3$.

step 2: $L^{-1} = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 0 & 9 \\ 0 & 3 & 0 \end{bmatrix}$. Therefore, $q_1 = \begin{bmatrix} 3 & 0 & 9 \end{bmatrix}$, $q_2 = \begin{bmatrix} 0 & 3 & 0 \end{bmatrix}$.

step 3: $Q = \begin{bmatrix} 3 & 0 & 9 \\ 4 & 0 & 18 \\ 0 & 3 & 0 \end{bmatrix}$. Using Q for another similarity transform.

$$\therefore \hat{A} = QA_cQ^{-1} = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right], \qquad \hat{B} = QB = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right]$$

Choose $\hat{F} = \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \in \mathbb{R}^{2\times 3}$.

$$\begin{aligned} \hat{A} + \hat{B}\hat{F} &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right] + \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right] \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \\ &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1+f_1 & 2+f_2 & 0 \\ \hline 0 & 0 & -3+f_3 \end{array}\right] \end{aligned}$$

If we choose $f_3 = 1$ then one of the eigenvalues of $\hat{A} + \hat{B}\hat{F}$ becomes -2. Next, we choose f_1 and f_2 such that the other two eigenvalues become -1,-1.

$\because (s+1)^2 = s^2 + 2s + 1, \quad 2 + f_2 = -2 \Rightarrow f_2 = -4$ and $-1 + f_1 = -1 \Rightarrow f_1 = 0$.

$$\begin{aligned} \therefore \hat{F} &= \begin{bmatrix} 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \therefore F_c &= \hat{F}Q = \begin{bmatrix} -16 & 0 & -72 \\ 0 & 3 & 0 \end{bmatrix} \end{aligned}$$

We can take $\hat{F} = \begin{bmatrix} F_c & M \end{bmatrix}$, where $M \in \mathbb{R}^{2\times 3}$ and can be choosen arbitrarily. We chose $M = 0$. The feedback matrix is, therefore,

$$F_1 = \hat{F}T^{-1} = \begin{bmatrix} 0 & 0 & 0 & 8 & -8 & 2 \\ 0 & 0 & 0 & -3 & 1 & 3/2 \end{bmatrix}$$

(one can check that $\sigma(A_1 + B_1F_1) = \{-1, -1, -1, -2, -2, -2\}$)
Using duality of controllability and observability, we have

$$L = F_1^T = \begin{bmatrix} 0 & 0 & 0 & 8 & -8 & 2 \\ 0 & 0 & 0 & -3 & 1 & 3/2 \end{bmatrix}^T \quad \text{and} \quad \sigma(A+LC) = \{-1, -1, -1, -2, -2, -2\}$$

Q3: Consider the system given by $4+4=8$

$$\begin{bmatrix}\dot{x_1}\\ \dot{x_2}\end{bmatrix} = \begin{bmatrix}-1 & 1\\ -1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u,\ y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}.$$

(a) Determine $x(0)$ so that for $u(t) = e^{-4t}, y(t) = ke^{-4t}$, where k is a real constant. Determine k for the present case.

Solution: Using the fact that $\dot{x_1} = x_1 + x_2$, $y = x_1 = ke^{-4t}$, and $u = e^{-4t}$, we find x_2 first.

$$\dot{x_2} = -x_1 + u \Rightarrow \dot{x_2} = -ke^{-4t} + e^{-4t} \Rightarrow x_2(t) = \frac{(k-1)}{4}e^{-4t} + c_1 \qquad (c_1 \text{ is a contant}).$$

Therefore, $\dot{x_1} = -ke^{-4t} + \dfrac{(k-1)}{4}e^{-4t} + c_1 = \dfrac{(-3k-1)}{4}e^{-4t} + c_1$

$$\therefore x_1(t) = \frac{(-3k-1)}{4}\frac{e^{-4t}}{-4} + c_1 t + c_2 = \frac{(3k+1)}{16}e^{-4t} + c_1 t + c_2 \qquad (c_2 \text{ is a contant}) \tag{3}$$

$\because y(t) = x_1(t)$, we have

$$ke^{-4t} = \frac{(3k+1)}{16}e^{-4t} + c_1 t + c_2 \Rightarrow c_1 = 0, c_2 = 0, \text{ and } \frac{(3k+1)}{16} = k.$$

Thus, $k = \dfrac{1}{13}$. Now,

$$x_1(0) = k = \frac{1}{13}, \qquad x_2(0) = \frac{(\frac{1}{13} - 1)}{4} = \frac{-3}{13} \qquad (\text{from } (2))$$

$\therefore$ Initial condition is $\begin{pmatrix}\frac{1}{13}\\ -\frac{3}{13}\end{pmatrix}$ and $k = \frac{1}{13}$.

(b) Let $u(t) = e^{\alpha t}$, Determine $x(0)$ that will result in $y(t) = ke^{\alpha t}$, Determine the conditions on α for this to be true. What is k in this case?

Solution: For $u(t) = e^{\alpha t}$ and $y(t) = ke^{\alpha t}$

$$x_2(t) = \frac{(1-k)}{\alpha}e^{\alpha t} + c_1$$

$$\dot{x_1} = -ke^{\alpha t} + \frac{(1-k)}{\alpha}e^{\alpha t} + c_1 = \left(\frac{-\alpha k + 1 - k}{\alpha}\right)e^{\alpha t} + c_1$$

$$\therefore x_1(t) = \left(\frac{1 - \alpha k - k}{\alpha}\right)\frac{e^{\alpha t}}{\alpha} + c_1 t + c_2 = ke^{\alpha t}$$

$$\therefore c_1 = 0, \qquad c_2 = 0, \qquad 1 - \alpha k - k = \alpha^2 k \Rightarrow k = \frac{1}{\alpha^2 + \alpha + 1}$$

$$\therefore x_1(0) = \frac{1}{\alpha^2 + \alpha + 1} \qquad \& \qquad x_2(0) = \frac{1}{\alpha}\left(1 - \frac{1}{\alpha^2 + \alpha + 1}\right) = \frac{\alpha + 1}{\alpha^2 + \alpha + 1}$$

Q4: Consider a system with the following transfer function:

$$G(s) = \begin{bmatrix}\frac{s-1}{s+1} & 0\\ \frac{s-1}{(s+1)^2} & \frac{s+2}{s^2-1}\end{bmatrix}$$

(a) Find the transmission zeros of the system.

(b) Find a minimal realization of the system and ascertain the invariant zeros of the system

(c) Find the poles of the system.

Solution: We find the McMillan form first.

$$G(s) = \begin{bmatrix} \dfrac{s-1}{s+1} & 0 \\ \dfrac{s-1}{(s+1)^2} & \dfrac{s+1}{s^2-1} \end{bmatrix} = \frac{1}{(s+1)^2(s-1)} \underbrace{\begin{bmatrix} (s-1)^2(s+1) & 0 \\ (s-1)^2 & (s+2)(s+1) \end{bmatrix}}_{P(s)}$$

$u(s)$	$P(s)$	$V(s)$	
$\begin{bmatrix} 1 & -s \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} s^3-s^2-s+1 & 0 \\ s^2-2s+1 & s^2+3s+2 \end{bmatrix}$		
	$\begin{bmatrix} s^2-2s+1 & 0 \\ s^2-2s+1 & s^2+3s+2 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$	$\begin{bmatrix} s^2-2s+1 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 5 & s \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} s^2-2s+1 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} -11s+5 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 5 & -11 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -11s+5 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} 36 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 1/36 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 36 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} 1 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 \\ 5s & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ -5s-1 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} 1 & 0 \\ -1 & s^2+3s+2 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ -1 & s^2+3s+2 \end{bmatrix}$		$\begin{bmatrix} 1 & 0 \\ 0 & s^2+3s+2 \end{bmatrix}$

$$G_{\mathtt{Mc}}(s) = \frac{1}{(s+1)^2(s-1)} \begin{bmatrix} 1 & 0 \\ 0 & s^2+3s+2 \end{bmatrix} = \begin{bmatrix} \dfrac{1}{(s+1)^2(s-1)} & 0 \\ 0 & \dfrac{s+2}{(s+1)(s-1)} \end{bmatrix}$$

a) Transmission zero of G(s):{-2}

b) We find the minimal representation next.

$$G(s) = \begin{bmatrix} \frac{s-1}{s+1} & 0 \\ \frac{s-1}{(s+1)^2} & \frac{s+2}{s^2-1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{s+1} \end{bmatrix}}_{G_1(s)} \underbrace{\begin{bmatrix} \frac{s-1}{s+1} & 0 \\ \frac{s-1}{s+1} & \frac{s+2}{s-1} \end{bmatrix}}_{G_2(s)}$$

We find a minimal realization for $G_1(s)$ first.

$$G_1(s) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{s+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{s+1} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_1 = -1, \qquad B_1 = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad C_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad D_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$G_2(s) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} \frac{-2}{s+1} & 0 \\ \frac{-2}{s+1} & \frac{3}{s-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \frac{1}{s+1} \begin{bmatrix} -2 & 0 \\ -2 & 0 \end{bmatrix} + \frac{1}{s-1} \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$$

$$A_2 = \left[\begin{array}{c|c} -1 & 0 \\ \hline 0 & 1 \end{array}\right], \qquad B_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad C_2 = \left[\begin{array}{c|c} -2 & 0 \\ -2 & 3 \end{array}\right], \qquad D_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$\therefore G_1(s)G_2(s)$ has the following realization:

$$\begin{aligned} A &= \begin{bmatrix} A_1 & B_1C_2 \\ 0 & A_2 \end{bmatrix} = \begin{bmatrix} -1 & -2 & 3 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} B_1D_2 \\ B_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \\ C &= \begin{bmatrix} C_1 & D_2C_2 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & 0 \end{bmatrix}, D = D_1D_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

$\because \texttt{rank}(\mathcal{C}_{AB}) = 3$ and $\texttt{rank}(\mathcal{O}_{CA}) = 3$, we have a minimal realization of $G(s)$: (A, B, C, D).
Invariant zeros: $\because (A, B, C, D)$ is a minimal realization, the transmission zeros and invariant zeros are same.
$\therefore -2$ is the invariant zero.

c) $\because$ Eigenvalues of A are $-1, -1, 1$ and (A, B, C, D) is minimal realization, poles are $\{-1, -1, 1\}$.

Q5: Form a matrix A with your Roll number in the following way:
For example: if your Roll number is 123456789, then the matrix will be of the form:

$$A := \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) Find the diagonal or the Jordan form (whichever is applicable) of A. Name this matrix J.

(b) Construct a matrix B with the minimum number of columns possible such that (J, B) is controllable.

(c) Construct a matrix C with the minimum number of rows possible such that (J, C) is observable.

Solution: This is unique for everyone. No particular answer.

EE656: Robust Control
Model answer Quiz 1

`Set - C`

Q1: Prove or disprove the following: $3 \times 5 = 15$

(a) If an observer exists, the system is observable.
False *(Detectability is a necessary and sufficient condition)*
Counter example to the statement:

$$\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

This is detectable. Consider $L = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$\sigma(A + LC) = \sigma\left(\begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right) = \sigma\left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \right) \subseteq \mathbb{C}^-$$

Thus, the observer -

$$\dot{\hat{x}} = A\hat{x} + Bu(t) + L(\hat{y} - y) \quad \text{and} \quad \hat{y} = C\hat{x} + Du(t)$$

will result in

$$\lim_{t \to \infty} [x(t) - \hat{x}(t)] = 0$$

(b) Every invariant zero is a transmission zero of the system.
Solution: ***False*** (This is true when (A,B,C,D) is a minimal realization of G(s))
e.g. $G(s) = \dfrac{s+2}{s+1}$ and consider $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = 1.$
You can verify that $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$
From $G(s) = \dfrac{s+2}{s+1}$, it is clear that $\{-2\}$ is a transmission zero.
Now, let $\lambda = -1$.

$$\texttt{rank} \begin{bmatrix} A - \lambda I & B \\ C & D \end{bmatrix} = \texttt{rank} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = 2 < \texttt{nrank} \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = 3$$

$\therefore -1$ is an invariant zero but not a transmission zero.

(c) The system (A, B) is controllable if and only if $(A + \alpha I, B)$ is controllable.
Solution: ***True***

If: (A+αI, B) controllable $\Rightarrow$ (A,B) controllable
We prove the contrapositive statement: If (A,B) uncontrollable $\Rightarrow$ (A+αI, B) uncontrollable.
$\because$ (A,B) uncontrollable. $\exists \mu \in \mathbb{C}$ and $\mu \in \sigma(A)$
such that

$$Av = \mu v \iff (A + \alpha I)v = (\mu + \alpha)v \tag{1}$$

Now conducting PBH test at $(\mu + \alpha)$ eigenvalue of (A+α I).

$$v^* \begin{bmatrix} (A + \alpha I) - (\mu + \alpha)I & B \end{bmatrix} = 0 \qquad \text{(using eq (1))}$$

$\therefore$ $(\mu + \alpha)$ is an uncontrollable eigenvalue of (A+α I).
$\therefore$ (A+α I,B) is uncontrollable.

only if: (A,B) controllable $\Rightarrow$ (A+αI, B) controllable.
We prove: (A+αI, B) uncontrollable $\Rightarrow$ (A,B) uncontrollable.
Let μ be an uncontrollable eigenvalue of $(A + \alpha I, B)$
and $(A + \alpha I)v = \mu v \Rightarrow Av = (\mu - \alpha)v$.
Hence PBH test on (A,B) at $\lambda = \mu - \alpha$ gives-

$$v^* \begin{bmatrix} A - (\mu - \alpha)I & B \end{bmatrix} = v^* \begin{bmatrix} (A + \alpha I) - \mu I & B \end{bmatrix} = 0$$

$\therefore$ (A,B) is uncontrollable.

(d) Given an uncontrollable system there exists an initial condition such that the system can be brought to rest in finite time.
Solution: ***False*** (Input needs to present for atleast some control)
e.g. $\dot{x} = x$: This is an uncontrollable system (more precisely autonomous). The states are $x(t) = e^t x(0)$. Clearly, $\nexists$ initial condition which can be brought to rest.

(e) $z_0 \in \mathbb{C}$ is a transmission zero of the system with transfer matrix $G(s)$ if and only if there exists a non-zero $u \in \mathbb{C}^{\mathtt{m}}$ such that

$$G(z_0)u = 0.$$

Solution: **False** (G(s) have to be full column normal rank)
For example, let

$$G(s) = \begin{bmatrix} \frac{1}{s+1} & \frac{1}{s+1} \\ \frac{1}{s+1} & \frac{1}{s+1} \end{bmatrix}.$$

Then,

$$G(s) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \qquad \forall s \in \mathbb{C}.$$

But all complex numbers cannot be transmission zeros.

Q2: Given the system 10

$$\dot{x} = \frac{1}{9} \begin{bmatrix} -6 & 0 & -30 & -15 & 18 & -21 \\ -11 & -15 & 86 & 13 & 54 & 11 \\ -6 & -9 & -30 & -6 & -18 & 6 \\ 30 & 18 & -3 & -33 & 63 & -30 \\ -1 & 6 & -2 & 11 & -27 & 1 \\ 6 & 9 & 3 & 6 & 18 & -33 \end{bmatrix} x \qquad y = \frac{1}{3} \begin{bmatrix} 2 & 0 & -5 & -1 & 0 & -2 \\ -1 & 0 & 10 & 2 & 3 & 4 \end{bmatrix} x.$$

Design an observer with poles at $\{-3,-3,-3,-4,-4,-4\}$. In case you are unable to place all the poles to the desired location, design the observer such that you can place as many poles as possible at the desired location.

Let $A_1 = A^T$ and $B_1 = C^T$. Note that $\texttt{rank}(\mathcal{C}_{A_1B_1}) = 3 < 6$. Therefore, the system is uncontrollable and dimension of controllable space is 3. We take a basis of the controllability space and extend it to $\mathbb{R}^6$.

On computation, we have $\mathcal{C}_{A_1B_1} = \langle \begin{bmatrix} 2 \\ 0 \\ -5 \\ -1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 10 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -8 \\ 3 \\ 29 \\ 7 \\ 9 \\ 8 \end{bmatrix} \rangle$.

We define $T = \begin{bmatrix} 2 & -1 & -8 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 1 & 0 \\ -5 & 10 & 29 & 0 & 0 & 1 \\ -1 & 2 & 7 & 0 & 0 & 0 \\ 0 & 3 & 9 & 0 & 0 & 0 \\ -2 & 4 & 8 & 0 & 0 & 0 \end{bmatrix}$. Using T as the similarity transformation matrix, we get

$$\hat{A} = T^{-1}AT = \left[\begin{array}{ccc|ccc} 0 & 0 & -3 & 17/6 & 17/6 & -1 \\ 0 & -3 & 0 & 7/6 & 7/6 & 1/3 \\ 1/3 & 0 & -2 & -1/6 & 5/18 & -1/3 \\ \hline 0 & 0 & 0 & -13/2 & -7/2 & -1 \\ 0 & 0 & 0 & 1/2 & -5/2 & 0 \\ 0 & 0 & 0 & 4 & 4 & -2 \end{array}\right] =: \left[\begin{array}{c|c} A_c & \hat{A}_{12} \\ \hline 0 & A_{uc} \end{array}\right],$$

$$\hat{B} = T^{-1}B = \left[\begin{array}{cc} 1/3 & 0 \\ 0 & 1/3 \\ 0 & 0 \\ \hline 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] =: \left[\begin{array}{c} B_c \\ \hline 0 \end{array}\right].$$

Note that $\sigma(A_c) = \{-1,-1,-3\}$ is the set of controllable eigenvalues and $\sigma(A_{uc}) = \{-4,-4,-3\}$ is the set of uncontrollable eigenvalues.

Desired eigenvalues: $\{-3,-3,-3,-4,-4,-4\}$

We just have to place the controllable modes at $\{-3,-3,-4\}$. Hence, we work with (A_c, B_c) next.

$$A_c = \begin{bmatrix} 0 & 0 & -3 \\ 0 & -3 & 0 \\ 1/3 & 0 & 2 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 1/3 & 0 \\ 0 & 1/3 \\ 0 & 0 \end{bmatrix} =: \begin{bmatrix} b_1 & b_2 \end{bmatrix}$$

Step 1: $L = \begin{bmatrix} b_1 & Ab_1 & b_2 \end{bmatrix} = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 0 & 1/3 \\ 0 & 1/9 & 0 \end{bmatrix}$. Here $d_1 = 2, \quad d_2 = 1, \quad \sigma_1 = 2, \quad \sigma_2 = 3$

Step 2: $L^{-1} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 9 \\ 0 & 3 & 0 \end{bmatrix}$. Therefore, $q_1 = \begin{bmatrix} 0 & 0 & 9 \end{bmatrix}$, $q_2 = \begin{bmatrix} 0 & 3 & 0 \end{bmatrix}$

step 3: $Q = \begin{bmatrix} 0 & 0 & 9 \\ 3 & 0 & -18 \\ 0 & 3 & 0 \end{bmatrix}$. Using Q for another similarity transform.

$$\therefore \hat{A} = QA_cQ^{-1} = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right], \qquad \hat{B} = QB = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right]$$

Choose $\hat{F} = \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \in \mathbb{R}^{2\times 3}$.

$$\begin{aligned} \hat{A} + \hat{B}\hat{F} &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 0 & 0 & -3 \end{array}\right] + \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ \hline 0 & 1 \end{array}\right] \begin{bmatrix} f_1 & f_2 & 0 \\ 0 & 0 & f_3 \end{bmatrix} \\ &= \left[\begin{array}{cc|c} 0 & 1 & 0 \\ -1+f_1 & 2+f_2 & 0 \\ \hline 0 & 0 & -3+f_3 \end{array}\right] \end{aligned}$$

If we choose $f_3 = 0$ then one of the eigenvalues of $\hat{A} + \hat{B}\hat{F}$ becomes -3. Next, we choose f_1 and f_2 such that the other two eigenvalues become -3,-4.

$\because (s+3)(s+4) = s^2 + 7s + 12$, $\quad -2 + f_2 = -7 \Rightarrow f_2 = -5$ and $-1 + f_1 = -12 \Rightarrow f_1 = -11$.

$$\begin{aligned} \therefore \hat{F} &= \begin{bmatrix} -11 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \therefore F_c &= \hat{F}Q = \begin{bmatrix} -15 & 0 & -9 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

We can take $\hat{F} = \begin{bmatrix} F_c & M \end{bmatrix}$, where $M \in \mathbb{R}^{2\times 3}$ and can be choosen arbitrarily. We chose $M = 0$. The feedback matrix is, therefore,

$$F_1 = \hat{F}T^{-1} = \begin{bmatrix} 0 & 0 & 0 & 7 & -10 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(one can check that $\sigma(A_1 + B_1F_1) = \{-3, -3, -3, -4, -4, -4\}$)
Using duality of controllability and observability, we have

$$L = F_1^T = \begin{bmatrix} 0 & 0 & 0 & 7 & -10 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \quad \text{and} \quad \sigma(A+LC) = \{-3, -3, -3, -4, -4, -4\}$$

Q3: Consider a system with the following transfer function:

$$G(s) = \begin{bmatrix} \frac{1}{s} & 0 \\ \frac{2}{s+1} & \frac{1}{s(s+1)} \end{bmatrix}$$

(a) Find the transmission zeros of the system.

(b) What is the normal rank of the $G(s)$?

(c) Find a minimal realization of the system and ascertain the invariant zeros of the system

(d) Find the poles of the system.

Solution: We find the McMillan form first.

$$G(s) = \begin{bmatrix} \frac{1}{s} & 0 \\ \frac{2}{s+1} & \frac{1}{s(s+1)} \end{bmatrix} = \frac{1}{s(s+1)} \underbrace{\begin{bmatrix} s+1 & 0 \\ 2s & 1 \end{bmatrix}}_{P(s)}$$

$u(s)$	$P(s)$	$V(s)$	
$\begin{bmatrix} 1 & -1/2 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} s+1 & 0 \\ 2s & 1 \end{bmatrix}$		$\begin{bmatrix} 1 & -1/2 \\ 2s & 1 \end{bmatrix}$
	$\begin{bmatrix} 1 & -1/2 \\ 2s & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 2s & s+1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 \\ -2s & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 2s & s+1 \end{bmatrix}$		$\begin{bmatrix} 1 & 0 \\ 0 & s+1 \end{bmatrix}$

$$G_{\mathtt{Mc}}(s) = \begin{bmatrix} \frac{1}{s(s+1)} & 0 \\ 0 & \frac{1}{s} \end{bmatrix}$$

a) The system has no transmission zeros.

b) $\mathtt{nrank}(G(s)) = 2$.

c) We find the minimal representation next.

$$G(s) = \begin{bmatrix} \frac{1}{s} & 0 \\ \frac{2}{s+1} & \frac{1}{s(s+1)} \end{bmatrix} = \frac{1}{s}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{s+1}\begin{bmatrix} 0 & 0 \\ 2 & -1 \end{bmatrix}$$

$$A = \left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \hline 0 & 0 & -1 \end{array}\right], \quad B = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ \hline 2 & -1 \end{array}\right], \quad C = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\because$ $\mathtt{rank}(\mathcal{C}_{AB}) = 3$ and $\mathtt{rank}(\mathcal{O}_{CA}) = 3$, we have a minimal realization of $G(s)$: (A, B, C, D).

Invariant zeros: $\because$ (A, B, C, D) is a minimal realization, the transmission zeros and invariant zeros are same.
$\therefore$ There are no invariant zeroes.

d) $\because$ Eigenvalues of A are $0, 0, -1$ and (A, B, C, D) is minimal realization, poles are $\{0, 0, -1\}$.

Q4: Consider the system given by $\qquad 4 + 4 = 8$

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \; y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

(a) Determine $x(0)$ so that for $u(t) = e^{-4t}, y(t) = ke^{-4t}$, where k is a real constant. Determine k for the present case.

Solution: Using the fact that $\dot{x_1} = x_1 + x_2$, $y = x_1 = ke^{-4t}$, and $u = e^{-4t}$, we find x_2 first.

$$\dot{x_2} = -x_1 + u \Rightarrow \dot{x_2} = -ke^{-4t} + e^{-4t} \Rightarrow x_2(t) = \frac{(k-1)}{4}e^{-4t} + c_1 \qquad (c_1 \text{ is a contant}).$$

Therefore, $\dot{x_1} = -ke^{-4t} + \dfrac{(k-1)}{4}e^{-4t} + c_1 = \dfrac{(-3k-1)}{4}e^{-4t} + c_1$

$$\therefore x_1(t) = \frac{(-3k-1)}{4}\frac{e^{-4t}}{-4} + c_1 t + c_2 = \frac{(3k+1)}{16}e^{-4t} + c_1 t + c_2 \qquad (c_2 \text{ is a contant}) \tag{2}$$

$\because y(t) = x_1(t)$, we have

$$ke^{-4t} = \frac{(3k+1)}{16}e^{-4t} + c_1 t + c_2 \Rightarrow c_1 = 0, c_2 = 0, \text{ and } \frac{(3k+1)}{16} = k.$$

Thus, $k = \dfrac{1}{13}$. Now,

$$x_1(0) = k = \frac{1}{13}, \qquad x_2(0) = \frac{(\frac{1}{13} - 1)}{4} = \frac{-3}{13} \qquad (\text{from } (2))$$

$\therefore$ Initial condition is $\begin{pmatrix} \frac{1}{13} \\ -\frac{3}{13} \end{pmatrix}$ and $k = \frac{1}{13}$.

(b) Let $u(t) = e^{\alpha t}$, Determine $x(0)$ that will result in $y(t) = ke^{\alpha t}$, Determine the conditions on α for this to be true. What is k in this case?

Solution: For $u(t) = e^{\alpha t}$ and $y(t) = ke^{\alpha t}$

$$x_2(t) = \frac{(1-k)}{\alpha}e^{\alpha t} + c_1$$

$$\dot{x_1} = -ke^{\alpha t} + \frac{(1-k)}{\alpha}e^{\alpha t} + c_1 = \left(\frac{-\alpha k + 1 - k}{\alpha}\right)e^{\alpha t} + c_1$$

$$\therefore x_1(t) = \left(\frac{1 - \alpha k - k}{\alpha}\right)\frac{e^{\alpha t}}{\alpha} + c_1 t + c_2 = ke^{\alpha t}$$

$$\therefore c_1 = 0, \qquad c_2 = 0, \qquad 1 - \alpha k - k = \alpha^2 k \Rightarrow k = \frac{1}{\alpha^2 + \alpha + 1}$$

$$\therefore x_1(0) = \frac{1}{\alpha^2 + \alpha + 1} \qquad \& \qquad x_2(0) = \frac{1}{\alpha}\left(1 - \frac{1}{\alpha^2 + \alpha + 1}\right) = \frac{\alpha + 1}{\alpha^2 + \alpha + 1}$$

Q5: Form a matrix A with your Roll number in the following way:
For example: if your Roll number is 123456789, then the matrix will be of the form:

$$A := \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) Find the diagonal or the Jordan form (whichever is applicable) of A. Name this matrix J.

(b) Construct a matrix B with the minimum number of columns possible such that (J, B) is controllable.

(c) Construct a matrix C with the minimum number of rows possible such that (J, C) is observable.

Solution: This is unique for everyone. No particular answer.